Product Master — Product Type — API Specification Design (API仕様設計)

Document control (改版履歴)

Field	Value
Document ID	004_DD-API-TYPE
Work item	WI-017
Based on	004_BD-TYPE v1 (DEC-007), 004_DB-TYPE v1 (DEC-008), 004_DD-TYPE v1 (DEC-009)
Created / updated	Claude (for ThanhTN), 2026-10-25

Version	Date	Author	Revision content
1	2026-10-25	Claude (for ThanhTN)	Initial creation: `type` on API-PM-01–05; FN-044/FN-045 processing

This document changes the five Product master operations of 004_DD-API. Common HTTP/JSON rules, authorization, other fields, status codes and error codes there (and WI-013's optional `pageSize`) are unchanged. No new route, error code or registry ID. API-PRD-01 (`GET /api/products`), the order APIs, the dashboard and the order CSV export are **not** changed (DEC-003).

Type value

`type` is a JSON **string**, exactly one of `FINISHED`, `SEMI_FINISHED`, `PART`, `RAW_MATERIAL` (case-sensitive; no trimming, no labels, no numbers). It maps to C# `ProductType` through a fixed code table, not `ToString()`, so enum member names can never leak into the contract.

Representation change

Response	Change
`ProductMasterItem`	`+ type` (string, always present, one of the four codes), placed after `name`
`PagedResult<ProductMasterItem>`	No change besides items

Example:

```
{ "id": "…", "sku": "P-1002", "name": "ブレーキディスクローター", "type": "PART", "unit": "枚",
  "drawingNumber": null, "isActive": true, "updatedAt": "2026-10-25T01:00:00Z", "version": 81234, "unitLocked": true }
```

API-PM-01 — `GET /api/product-master` (added query field)

Query field	Required	Rule / default
`type`	no	Absent or empty = all types. Otherwise one of the four codes; any other value → 400 `VALIDATION`, `errors.type`

Applied with `q` and `state` before count and paging (AND). Ordering unchanged (SKU, then ID). `page` / `pageSize` behaviour unchanged.

API-PM-02 — `GET /api/product-master/{id}`

Returns `type`. No other change.

API-PM-03 — `POST /api/product-master` (added body field)

Body field	Required	Validation
`type`	yes	One of the four codes; missing, `null`, non-string or unknown → 400 `VALIDATION`, `errors.type`

201 response includes `type`.

API-PM-04 — `PUT /api/product-master/{id}` (added body field)

Body field	Required	Rule
`type`	yes	As create. May change on any product, active or retired, referenced by orders or not; no lock, no new conflict

Full replacement as today. Validation order: field validation (including `type`) → version (409 `PRODUCT_STALE`) → unit lock (409 `PRODUCT_UNIT_LOCKED`), unchanged. A `type`-only change still bumps `version` and `updatedAt`.

API-PM-05 — `POST /api/product-master/{id}/retire`

Body unchanged (`version` only; `type` is rejected as an unknown property, as any other). Response includes `type`, unchanged by retirement.

Processing (FN-044, FN-045)

Step	Where	Rule
Parse list `type`	`ProductMasterListQuery.TryCreate`	`null` / `""` → no filter; known code → `ProductType`; else `errors["type"] = ["VALIDATION"]` alongside other query errors
Filter	`ProductMasterRepository` list	`products.Where(p => p.Type == query.Type)` when set, before count
Validate write	`ProductMasterService.Validate`	`type` code → `ProductType`; failure adds `errors["type"]`, reported together with other field errors
Save	Create: set `Type`. Update: assign `Type` under the existing row lock and version check	No extra query; `updated_at_utc` set as for other edits
Map out	`ProductMasterItem` factory	`Type` → code
DB	004_DB-TYPE	`NOT NULL` + `ck_products_product_type` back the API check

Telemetry: unchanged outcome counters; the type value is not logged (not sensitive, but no need).

Error mapping (additions only)

Condition	HTTP / code	Field / client action
Missing/unknown `type` on create or update	400 `VALIDATION`	`errors.type` → MSG-E026 at the 「種別」 field; draft kept
Unknown list `type`	400 `VALIDATION`	`errors.type`; the screen never sends one (004_DD-TYPE URL state)

Compatibility and rollout

- **Breaking for writers:** create/update now require `type`. The only clients are this frontend, its tests and E2E fixtures; they change in the same release. The migration makes the column `NOT NULL`, so API and migration deploy together (004_DB-TYPE).
- **Readers:** `type` is an added response field; clients ignoring unknown fields are unaffected.

- **Order and other product APIs:** unchanged, so order screens, dashboard and CSV keep working without change.

Test viewpoints

Level	Cases
Unit	`TryCreate` : absent, `""`, each code, `part` (wrong case), `X` , combined with bad `page` (both errors). `Validate` : missing, `null` , `123` , unknown, each code
Integration	POST each code → 201 with `type` ; POST without `type` → 400 `errors.type` ; PUT type change on a unit-locked retired product → 200, new version; PUT stale version with a bad type → 400 first; list `type=PART` → only parts, `total` counts filtered rows, combined with `q` , `state=retired` , `pageSize=10` ; list `type=foo` → 400; retire body with `type` → 400